

Question_4:

1. Verilog code :

- Module code:

```
1 module spr(clk,rst,din,addr,wr_en,rd_en,blk_select,addr_en,dout_en,dout,parity_out);
2 parameter MEM_WIDTH=16;
3 parameter MEM_DEPTH=1024;
4 parameter ADDR_SIZE=10;
5 parameter ADDR_PIPELINE="FALSE";
6 parameter DOUT_PIPELINE="TRUE";
7 parameter PARITY_ENABLE=1;
8 //////////////////////////////////////////////////
9 input clk,rst;
10 input rd_en,wr_en,blk_select,addr_en,dout_en;
11 input [MEM_WIDTH-1:0] din;
12 input [ADDR_SIZE-1:0] addr;
13 output reg parity_out;
14 output [MEM_WIDTH-1:0] dout;
15 //////////////////////////////////////////////////
16 reg [MEM_WIDTH-1:0] mem[MEM_DEPTH-1:0] ;
17 //////////////////////////////////////////////////
18 reg [ADDR_SIZE-1:0] addr_ff;
19 wire [ADDR_SIZE-1:0] addr_mem;
20 reg [MEM_WIDTH-1:0] dout_ff;
21 reg [MEM_WIDTH-1:0] dout_mem;
22 //////////////////////////////////////////////////
23 assign addr_mem=(ADDR_PIPELINE=="FALSE")?addr:addr_ff;
24 assign dout=(DOUT_PIPELINE=="FALSE")?dout_mem:dout_ff;
25 always @(posedge clk) begin
26 if(rst)
27 addr_ff<=0;
28 else if((addr_en))
29 begin
30 addr_ff<=addr;
31 end
32 end
33
34 always @(posedge clk ) begin
35 if(rst)
36 dout_ff<=0;
37 else if (dout_en)begin
38 dout_ff<=dout_mem;
39 end
40 end
41 always @(posedge clk) begin
42 if(rst) begin
43 dout_mem <= 0;
44 parity_out<=0;
45 end
46 else begin
47 if (blk_select) begin
48 if (wr_en) begin
49 mem[addr_mem] <= din;
50 end
51 if (rd_en) begin
52 dout_mem <= mem[addr_mem];
53 end
54
55 if(PARITY_ENABLE==1) begin
56 parity_out<=~mem[addr_mem];
57 end
58 end
59 end
60 end
61 endmodule
```

- Test bench:

```

1  module spr_tb();
2  parameter MEM_WIDTH=16;
3  parameter MEM_DEPTH=1024;
4  parameter ADDR_SIZE=10;
5  reg clk,rst;
6  reg rd_en,wr_en,blk_select,addr_en,dout_en;
7  reg [MEM_WIDTH-1:0] din;
8  reg [ADDR_SIZE-1:0] addr;
9  wire parity_out;
10 wire [MEM_WIDTH-1:0] dout;
11 //////////////////////////////////////////////////
12 spr SPR (clk,rst,din,addr,wr_en,rd_en,blk_select,addr_en,dout_en,dout,parity_out);
13 //////////////////////////////////////////////////
14 initial begin
15     clk=0;
16     forever
17         #1 clk=~clk;
18 end
19 integer i;
20 initial begin
21     $readmemh("mem.dat",SPR.mem);
22     rst=1;
23     rd_en=0;
24     wr_en=1;
25     dout_en=0;
26     blk_select=1;
27     addr=10'd1020;
28     din=16'd20;
29     addr_en=0;
30     @(negedge clk);
31     rst=0;
32
33     rd_en=0;
34
35     for(i=0;i<30;i=i+1)
36     begin
37         blk_select=$random;
38
39         addr=$random;
40         wr_en=$random;
41         addr_en=$random;
42         dout_en=$random;
43         din=$random;
44         @(negedge clk);
45     end
46     //////////////////////////////////////////////////
47     wr_en=0;
48
49
50     for(i=0;i<30;i=i+1) begin
51         blk_select=$random;
52         addr=$random;
53         rd_en=$random;
54         addr_en=$random;
55         dout_en=$random;
56
57         @(negedge clk);
58     end
59     $stop;
60 end
61 endmodule

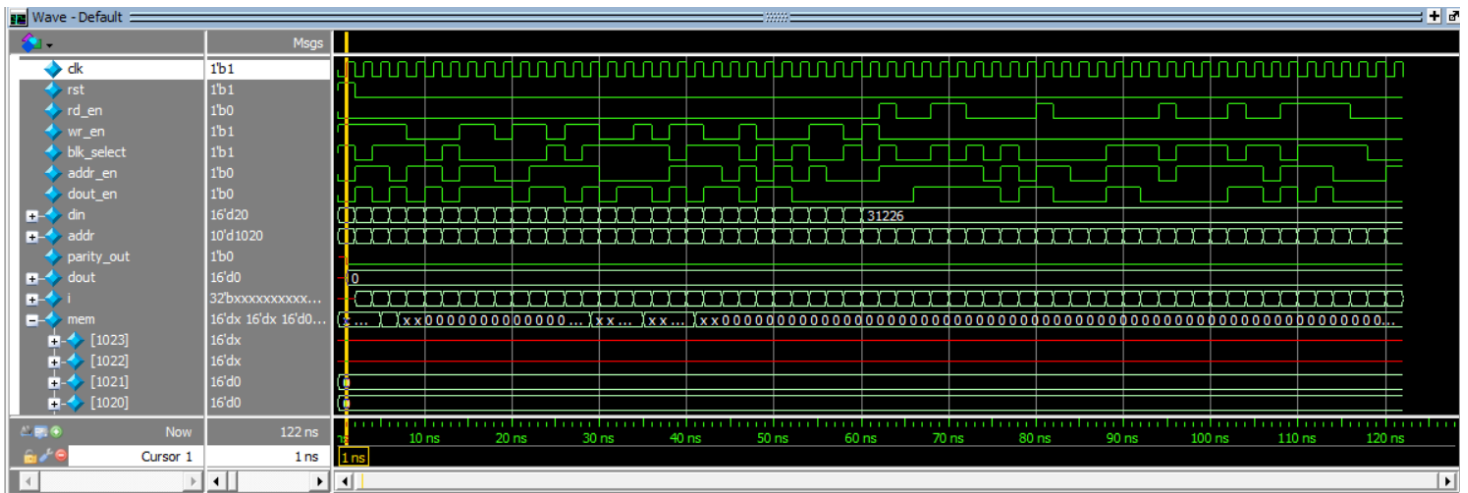
```

- Do file:

```
vlib work
vlog Q4_1.v Q4_1_tb.v
vsim -voptargs=+acc work.spr_tb
add wave *
run -all
#quit -sim
```

2. Waveform snippet:

- Waveform:



3. Synthesis tool:

• Elaboration:

The screenshot displays the Vivado IDE during the Elaboration stage. The Sources panel on the left shows the project structure with 'spr (Q4_1.v)' selected. The Schematic panel on the right shows a detailed logic diagram with components like RTL_MUX, RTL_AND, mem_reg, and RTL_RAM. The Messages panel at the bottom shows synthesis warnings and info messages.

• Synthesis :

The screenshot displays the Vivado IDE during the Synthesis stage. The Sources panel on the left shows the project structure with 'spr' selected. The Schematic panel on the right shows a high-level block diagram of the design. The Messages panel at the bottom shows synthesis warnings and info messages.

Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	F8 Muxes (8150)	Bonded IOB (106)	BUFGCTRL (32)
spr	280	33	128	64	49	1

Design Timing Summary					
Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	5.335 ns	Worst Hold Slack (WHS):	0.154 ns	Worst Pulse Width Slack (WPWS):	3.750 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	33	Total Number of Endpoints:	33	Total Number of Endpoints:	290
All user specified timing constraints are met.					

- Netlist:

```

12 // -----
13 `timescale 1 ps / 1 ps
14
15 (* ADDR_PIPELINE = "FALSE" *) (* ADDR_SIZE = "10" *) (* DOUT_PIPELINE = "TRUE" *)
16 (* MEM_DEPTH = "1024" *) (* MEM_WIDTH = "16" *) (* PARITY_ENABLE = "1" *)
17 (* STRUCTURAL_NETLIST = "yes" *)
18 module spr
19     (clk,
20      rst,
21      din,
22      addr,
23      wr_en,
24      rd_en,
25      blk_select,
26      addr_en,
27      dout_en,
28      dout,
29      parity_out);
30     input clk;
31     input rst;
32     input [15:0]din;
33     input [9:0]addr;
34     input wr_en;
35     input rd_en;
36     input blk_select;
37     input addr_en;
38     input dout_en;
39     output [15:0]dout;
40     output parity_out;
41
42     wire [9:0]addr;
43     wire [9:0]addr_IBUF;
44     wire blk_select;
45     wire blk_select_IBUF;
46     wire clk;
47     wire clk_IBUF;
48     wire clk_IBUF_BUF;
49     wire [15:0]din;
50     wire [15:0]din_IBUF;
51     wire [15:0]dout;
52     wire [15:0]dout_OBUF;
53     wire dout_en;
54     wire dout_en_IBUF;
55     wire [15:0]dout_mem;
56     wire [15:0]dout_mem0;
57     wire \dout_mem[15]_3_1_n_0;
58     wire mem_reg_0_255_0_0_1_1_n_0;
59     wire mem_reg_0_255_0_0_n_0;
60     wire mem_reg_0_255_10_10_n_0;
61     wire mem_reg_0_255_11_11_n_0;
62     wire mem_reg_0_255_12_12_n_0;
63     wire mem_reg_0_255_13_13_n_0;
64     wire mem_reg_0_255_14_14_n_0;
65     wire mem_reg_0_255_15_15_n_0;
66     wire mem_reg_0_255_1_1_n_0;
67     wire mem_reg_0_255_2_2_n_0;
68     wire mem_reg_0_255_3_3_n_0;
69     wire mem_reg_0_255_4_4_n_0;
70     wire mem_reg_0_255_5_5_n_0;
71     wire mem_reg_0_255_6_6_n_0;
72     wire mem_reg_0_255_7_7_n_0;
73     wire mem_reg_0_255_8_8_n_0;
74     wire mem_reg_0_255_9_9_n_0;
75     wire mem_reg_256_511_0_0_1_1_n_0;
76     wire mem_reg_256_511_0_0_n_0;
77     wire mem_reg_256_511_10_10_n_0;
78     wire mem_reg_256_511_11_11_n_0;
79     wire mem_reg_256_511_12_12_n_0;
80     wire mem_reg_256_511_13_13_n_0;
81     wire mem_reg_256_511_14_14_n_0;
82     wire mem_reg_256_511_15_15_n_0;

```

```

83 wire mem_reg_256_511_1_1_n_0;
84 wire mem_reg_256_511_2_2_n_0;
85 wire mem_reg_256_511_3_3_n_0;
86 wire mem_reg_256_511_4_4_n_0;
87 wire mem_reg_256_511_5_5_n_0;
88 wire mem_reg_256_511_6_6_n_0;
89 wire mem_reg_256_511_7_7_n_0;
90 wire mem_reg_256_511_8_8_n_0;
91 wire mem_reg_256_511_9_9_n_0;
92 wire mem_reg_512_767_0_0_i_1_n_0;
93 wire mem_reg_512_767_0_0_n_0;
94 wire mem_reg_512_767_10_10_n_0;
95 wire mem_reg_512_767_11_11_n_0;
96 wire mem_reg_512_767_12_12_n_0;
97 wire mem_reg_512_767_13_13_n_0;
98 wire mem_reg_512_767_14_14_n_0;
99 wire mem_reg_512_767_15_15_n_0;
100 wire mem_reg_512_767_1_1_n_0;
101 wire mem_reg_512_767_2_2_n_0;
102 wire mem_reg_512_767_3_3_n_0;
103 wire mem_reg_512_767_4_4_n_0;
104 wire mem_reg_512_767_5_5_n_0;
105 wire mem_reg_512_767_6_6_n_0;
106 wire mem_reg_512_767_7_7_n_0;
107 wire mem_reg_512_767_8_8_n_0;
108 wire mem_reg_512_767_9_9_n_0;
109 wire mem_reg_768_1023_0_0_i_1_n_0;
110 wire mem_reg_768_1023_0_0_n_0;
111 wire mem_reg_768_1023_10_10_n_0;
112 wire mem_reg_768_1023_11_11_n_0;
113 wire mem_reg_768_1023_12_12_n_0;
114 wire mem_reg_768_1023_13_13_n_0;
115 wire mem_reg_768_1023_14_14_n_0;
116 wire mem_reg_768_1023_15_15_n_0;
117 wire mem_reg_768_1023_1_1_n_0;

```

- Constrains file:

```

179 create_debug_port u_ila_0 probe
180 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe2]
181 set_property port_width 16 [get_debug_ports u_ila_0/probe2]
182 connect_debug_port u_ila_0/probe2 [get_nets [list {din_IBUF[0]} {din_IBUF[1]} {din_IBUF[2]} {din_IBUF[3]} {din_IBUF[4]} {din_IBUF[5]} {din_IBUF[6]} {din_IBUF[7]} {din_IBUF[8]} {din_IBUF[9]} {din_IBUF[10]} {din_IBUF[11]} {din_IBUF[12]} {din_IBUF[13]} {din_IBUF[14]} {din_IBUF[15]}]]
183 create_debug_port u_ila_0 probe
184 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe3]
185 set_property port_width 1 [get_debug_ports u_ila_0/probe3]
186 connect_debug_port u_ila_0/probe3 [get_nets [list blk_select_IBUF]]
187 create_debug_port u_ila_0 probe
188 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe4]
189 set_property port_width 1 [get_debug_ports u_ila_0/probe4]
190 connect_debug_port u_ila_0/probe4 [get_nets [list clk_IBUF]]
191 create_debug_port u_ila_0 probe
192 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe5]
193 set_property port_width 1 [get_debug_ports u_ila_0/probe5]
194 connect_debug_port u_ila_0/probe5 [get_nets [list dout_en_IBUF]]
195 create_debug_port u_ila_0 probe
196 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe6]
197 set_property port_width 1 [get_debug_ports u_ila_0/probe6]
198 connect_debug_port u_ila_0/probe6 [get_nets [list parity_out_OBUF]]
199 create_debug_port u_ila_0 probe
200 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe7]
201 set_property port_width 1 [get_debug_ports u_ila_0/probe7]
202 connect_debug_port u_ila_0/probe7 [get_nets [list rd_en_IBUF]]
203 create_debug_port u_ila_0 probe
204 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe8]
205 set_property port_width 1 [get_debug_ports u_ila_0/probe8]
206 connect_debug_port u_ila_0/probe8 [get_nets [list rst_IBUF]]
207 create_debug_port u_ila_0 probe
208 set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe9]
209 set_property port_width 1 [get_debug_ports u_ila_0/probe9]
210 connect_debug_port u_ila_0/probe9 [get_nets [list wr_en_IBUF]]
211 set_property C_CLK_INPUT_FREQ_HZ 300000000 [get_debug_cores dbg_hub]
212 set_property C_ENABLE_CLK_DIVIDER false [get_debug_cores dbg_hub]
213 set_property C_USER_SCAN_CHAIN 1 [get_debug_cores dbg_hub]

```

• Implementation:

The screenshot displays the Vivado IDE interface. The left pane shows the 'Netlist' view with a hierarchical tree structure: 'spr' (Nets: 257) contains 'Leaf Cells' (171), which includes 'dbg_hub' (dbg_hub) and 'u_ila_0' (u_ila_0). The right pane shows the 'Device' view, a 2D grid representing the device layout with components labeled X0Y2, X1Y2, X0Y1, and X1Y1.

The screenshot shows the 'Messages' window in Vivado. It contains two main sections: 'Vivado Commands' and 'Synthesis'. The 'Synthesis' section shows a warning message: 'All user specified timing constraints are met.'.

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 2.663 ns	Worst Hold Slack (WHS): 0.030 ns	Worst Pulse Width Slack (WPWS): 3.750 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 4621	Total Number of Endpoints: 4605	Total Number of Endpoints: 2956
All user specified timing constraints are met.		

Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	F8 Muxes (8150)	Slice (8150)	LUT as Logic (20800)	LUT as Memory (9600)	LUT Flip Flop Pairs (20800)	Block RAM Tile (50)	Bonded IOB (106)
▼ N spr	1787	2392	143	64	796	1361	426	939	1.5	49
> ▼ dbg_hub (dbg_hub)	476	727	0	0	231	452	24	316	0	0
> ▼ u_ila_0 (u_ila_0)	1031	1632	15	0	489	885	146	615	1.5	0